

The $S_5/\mathbb{F}_5$ No-Go as an Instance of James's Submodule Theorem: A Trichotomy of Obstruction, Possibility, and Existence for Group-Invariant Codes

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Abstract

The Rocq formalization `reconstruct/s5_nogo.v` proves, by an elementary difference-vector argument, that the natural permutation module $\mathbb{F}_5^5$ of the symmetric group S_5 has invariant submodules of dimension only $0, 1, 4, 5$, so no secret-encoding invariant code of the gap-window dimensions 3 or 4 exists. This note records that the result is not a single witness but the smallest instance of a named general rule. The obstruction is governed by James's Submodule Theorem for Young permutation modules. For the natural S_n module the realizable submodule dimensions are $\{0, 1, n-1, n\}$ in every characteristic, since the difference-vector argument is characteristic-free and the standard module is the irreducible piece of dimension $n-1$, so the S_5 gap window $\{3, 4\}$ is infeasible over every field. The divisibility condition $p \mid n$ controls only whether the structure is uniserial or split, not the available dimensions, so the obstruction is characteristic-independent and Maschke does not remove it. Algebraic-geometry-code secret sharing with automorphism-transfer supplies a constructive existence rule for suitable groups, with the threshold gap governed by the genus rather than by a submodule lattice, but for the natural S_5 action that route is obstructed as well.

1 The question and the verdict

A code-based (Massey) secret-sharing scheme that recovers a secret needs a G -invariant submodule W , a linear code, of a representation module over a field F , of a dimension forced into a threshold gap window. The $S_5/\mathbb{F}_5$ instance acts on $\mathbb{F}_5^5$ by the natural permutation module, with the secret in a fixed coordinate adjoined to the five shuffled shares. The formalization establishes that the available invariant submodule dimensions are exactly $\{0, 1, 4, 5\}$, so the intermediate dimensions 2 and 3 are absent and the gap-window code cannot be built.

The verdict of the present survey is that this is a rule, and the rule has a name. The hand-proof in `s5_nogo.v` re-derives the same $\{0, 1, 4, 5\}$ lattice that James's Submodule Theorem forces. Each of the three tiers below, obstruction, possibility, and constructive existence, corresponds to a named theorem.

*Literature synthesis assembled from a fan-out, adversarially verified deep-research pass over the modular representation theory of symmetric groups and the algebraic-geometry-code secret-sharing literature. The classical core is documented through graduate lecture notes that quote the named theorems verbatim. Bibliographic details should be confirmed against the authoritative published sources before formal citation.

2 Tier 1: the obstruction rule (which groups cannot)

The available invariant-code dimensions are precisely the submodule dimensions of the permutation module, and those are controlled by a single characteristic-free theorem.

James’s Submodule Theorem [10, 8]. For any field F and any partition λ of n , every FS_n -submodule U of the Young permutation module M^λ either contains the Specht module S^λ or is contained in its orthogonal complement $(S^\lambda)^\perp$ under the tabloid bilinear form. No submodule lies strictly between S^λ and $(S^\lambda)^\perp$.

For the natural module $M^{(n-1,1)}$, the relevant object for S_n acting on $\mathbb{F}_p^n$, one has $\dim M = n$, $\dim S^{(n-1,1)} = n - 1$ for the sum-zero subspace P_0 , and $\dim(S^{(n-1,1)})^\perp = 1$ for the all-ones line.

What the divisibility condition controls. The Specht module $S^{(n-1,1)}$ is irreducible if and only if $\text{char } F = 0$ or $\text{char } F \nmid n$ [10]. This is Peel’s theorem on hook partitions: $S^{(n-1,1)}$ has exactly two composition factors when $p \mid n$ for odd p , and is irreducible otherwise. When $p \mid n$, the all-ones vector has coordinate sum $n \equiv 0$, hence lies inside the sum-zero space, so $(S^{(n-1,1)})^\perp \subset S^{(n-1,1)}$ and the lattice collapses to the uniserial chain

$$0 \subsetneq \langle \mathbf{1} \rangle \subsetneq P_0 = S^{(n-1,1)} \subsetneq M^{(n-1,1)}, \quad \dim \in \{0, 1, n-1, n\}.$$

This is the $\{0, 1, 4, 5\}$ chain at $n = 5$, and the rigidity that any nonconstant vector forces the dimension from at most 1 straight to at least $n-1$. This rigidity is characteristic-free: the difference-vector argument uses no field hypothesis, so the submodule dimensions are $\{0, 1, n-1, n\}$ over every field. The divisibility $p \mid n$ governs only whether $\langle \mathbf{1} \rangle$ lies inside P_0 , that is uniserial versus split, not which dimensions occur. The simple modules are fixed as well: in prime characteristic p the simple FS_n -modules are $D^\lambda = S^\lambda / (S^\lambda \cap (S^\lambda)^\perp)$ for p -regular λ , James’s 1976 classification recorded in [10].

Naming the $S_5/\mathbb{F}_5$ module precisely. The module is the natural permutation module $M^{(n-1,1)}$, and the precise modular phenomenon is that when $\text{char } F \mid n$ the $(n-1, 1)$ -Specht module is a submodule but not a direct summand of $M^{(n-1,1)}$ [8]. Maschke’s complement fails exactly there, and the intermediate piece does not split off.

Generalization. The no-go holds for S_n on $\mathbb{F}_p^n$ exactly when $p \mid n$, and then the entire window of intermediate dimensions $2, 3, \dots, n-2$ is forbidden, not merely a single gap. The case $n = p = 5$ is the smallest instance. The transposition-difference mechanism of `s5_nogo.v`, in which 2-transitivity breeds every difference vector $e_i - e_j$ and spans the dimension- $(n-1)$ sum-zero space, is the concrete shadow of this theorem.

3 Tier 2: the possibility rule (which groups can)

Maschke’s theorem [8]. If $\text{char } F \nmid |G|$, every submodule has a G -invariant complement, the module is semisimple, and every partial sum of irreducible-constituent dimensions is realized as a submodule. This is the general possibility rule. Whether it returns the gap-window dimensions depends on the irreducible-constituent dimensions of the particular module.

The possibility rule does not rescue S_5 (correction). For the natural module $M^{(n-1,1)}$ the constituents are the trivial module of dimension 1 and the standard module of dimension $n - 1$, so the realizable submodule dimensions are $\{0, 1, n - 1, n\}$ in every characteristic. Maschke changes the structure from uniserial to split, but it does not change this dimension profile. The symmetric group S_5 has no 3-dimensional ordinary irreducible at all, its irreducible dimensions being $\{1, 1, 4, 4, 5, 5, 6\}$, and its standard module is the irreducible of dimension 4. The dimension-3 object $D^{(4,1)}$, the heart, exists only in characteristic 5, and even there only as the subquotient $P_0/\langle \mathbf{1} \rangle$, never as a submodule. The difference-vector argument of `s5_nogo.v` is itself characteristic-free, so the $\{3, 4\}$ gap-window infeasibility for the natural S_5 action holds over every field. An earlier reading, following a comment in the Rocq source, claimed that coprime characteristic restores dimensions 2 and 3. That is wrong for S_5 and is corrected here. The contrast is S_4 , whose standard module is the irreducible of dimension 3, so a dimension-3 invariant code does exist there, which is exactly why the no-g needs five shuffled points rather than four.

Refinement away from the corner [9]. For $n \geq 66$ and $p > 3$, the multiplicity-free permutation characters of S_n are completely classified, and each summand of the modular permutation module is self-dual and carries a Specht filtration whose factor dimensions are read off the ordinary character. The Specht module S^λ appears in the filtration if and only if the ordinary character χ^λ is a constituent of the permutation character. Outside small or defining characteristic, the modular permutation module retains a clean and predictable composition series.

Small characteristic genuinely breaks order [9]. The hypotheses $p > 3$ and large n are necessary, not artifacts. In characteristic 3 the module $F[S_9/PTL(2, 8)]$ has no Specht filtration at all, and in characteristic 2 the module $\mathbb{F}_2[S_6/(S_3 \wr S_2)]$ has a Specht filtration but not one with the factors predicted by its ordinary character. The $S_5/\mathbb{F}_5$ uniserial collapse is the cleanest member of this small-and-defining-characteristic family.

The semisimple multiplicity data [5, 4]. The composition-factor multiplicities underlying the realizable submodule dimensions in the Maschke regime are Littlewood-Richardson coefficients: over a field where the skew Specht module is semisimple, $V^{\lambda/\mu} \cong \bigoplus_\nu (V^\nu)^{c_{\mu\nu}^\lambda}$. The James-Peel contribution is the corresponding Specht filtration with the same multiplicities valid over an arbitrary field, the direct-sum form holding only in the semisimple regime.

4 Tier 3: the constructive rule (which groups certainly do)

The constructive engine abandons the permutation-module lattice and switches to genus-controlled constructions, in which the threshold gap is a different object.

The AG-LSSS gap is the genus [2]. From a residue code on a genus- g curve, subsets of size below $n - m$ are unqualified and subsets of size at least $n - m + 2g$ are qualified, so the privacy and reconstruction thresholds are separated by a gap of width exactly $2g$, with $m = \deg G$. The thresholds are bracketed by the minimum and dual distances, $n - m \leq d_{\min} \leq d_{\text{cheat}} \leq n - m + 2g$. Genus 0 recovers Reed-Solomon and hence Shamir, with gap 0. The gap is driven by the genus, not by a submodule lattice.

Existence over any fixed field [1]. For every fixed finite field there is an infinite family of ideal linear secret-sharing schemes that is asymptotically good with strong multiplication, built on algebraic-geometry codes over good towers of function fields. This is an unconditional possibility result, and it imposes no group symmetry on its own.

The symmetry-transfer rule that guarantees invariance [6]. If a divisor D is very ample and the evaluation point set E is large enough, namely $n > r \cdot \deg D$ for some $r \geq 1$, then the curve-automorphism group fixing D and E is realized exactly as the permutation automorphism group of the code, and the two groups are isomorphic, $\text{Aut}_{D,E}(X) \cong \text{Perm}(C)$. Concretely, for a smooth projective curve of genus $g \geq 2$ with $\deg D \geq 2g + 1$ and $|E| \geq (1 + g) \deg D$, the code has $n = \deg E$, $k = \deg D + 1 - g$, $d \geq \deg E - \deg D$, and its permutation automorphism group is the prescribed curve symmetry. Choosing a curve with the desired automorphism group, such as the Hermitian, Suzuki, Ree, or modular curves with their large automorphism groups, transfers that symmetry to the code.

Group-algebra codes [7, 3]. The Berman codes and their duals, derived from the group algebra $\mathbb{F}_2[\mathbb{Z}_n^m]$, have automorphism group the wreath product $S_n \wr S_m$ in product action, with the related abelian codes realizing the same group for odd $n \geq 5$. A sharp modular-versus-semisimple existence dichotomy parallels the no-go: assuming the MDS conjecture, which is a theorem for prime fields, the only nontrivial MDS group codes in a non-semisimple group algebra $\mathbb{F}_p G$ occur for $G = C_p$ with p odd, where they coincide with extended Reed-Solomon codes, and outside this cyclic corner an MDS group code forces $\mathbb{F}_p G$ to be semisimple.

5 Location of the $S_5/\mathbb{F}_5$ case

The case sits in the multiplicity-free corner of James’s Submodule Theorem, with $n = p = 5$. For the natural S_n action the realizable submodule dimensions are $\{0, 1, n - 1, n\}$ in every characteristic, because the difference-vector argument is characteristic-free and the standard module is the irreducible piece of dimension $n - 1$. The divisibility condition $p \mid n$ controls only the module structure, uniserial when $p \mid n$ and split when $p \nmid n$. It does not change the available dimensions. So the S_5 gap window $\{3, 4\}$ is infeasible over every field, Maschke does not restore it, and the no-go is characteristic-independent for the natural action. The same holds for S_n on $\mathbb{F}_p^n$ for every n , the forbidden interior being the dimensions $2, \dots, n - 2$.

6 Caveats and refuted framings

Three points constrain how the result should be stated.

- The no-go should not be framed as a distributive-lattice, join-irreducible-poset, or Ext-quiver result. Several such framings were tested and refuted. The supported mechanism is specifically James’s Submodule Theorem together with the $(n - 1, 1)$ irreducibility dichotomy, and nothing more exotic.
- The two notions of gap are different objects. The algebraic-geometry threshold gap is the genus window of width $2g$, a curve and code parameter. The representation-theoretic gap is the set of missing submodule dimensions. The constructive results show how to build symmetric codes with controlled parameters. They do not show the submodule obstruction reappearing inside

algebraic-geometry codes. The two are independent constraints that a recoverable invariant scheme must both satisfy.

- Source quality varies. The James, Maschke, and Peel core is textbook-standard and documented through lecture notes that quote the theorems verbatim, hence high confidence. The Berman automorphism result is a single recent preprint, hence medium confidence. The MDS group-code dichotomy is conditional on the MDS conjecture, discharged for prime fields.
- For the specific natural-action S_5 protocol of `pgg-smc`, the Tier-3 constructive routes were evaluated and rejected. Genus 0 is excluded because $|S_5| = 120$ exceeds the Klein finite-subgroup ceiling 60, Wiman’s classification rules out genus 3, and Bring’s curve at genus 4 is only axiomatized while the algebraic-geometry recovery route is excluded by the no-go itself. Recovery survives only for the small cyclic instances and the intransitive product. A companion note records the rejected Bring-curve and genus-0 PGL routes with commit and file pointers.

7 Open problems

1. A clean criterion for which other 2-transitive groups, such as projective special linear, affine, or sporadic 2-transitive groups, carry the defining-characteristic obstruction on their natural modules. The 2-transitivity is what breeds the difference vectors, but no consulted source pins down the general cross-versus-defining split for an arbitrary 2-transitive group.
2. Whether a nontrivial group symmetry can be imposed on the asymptotically good ideal scheme family while preserving asymptotic goodness and strong multiplication. The automorphism-transfer theorem supplies the mechanism and the asymptotically good family exists, but no consulted source combines them into an asymptotically good symmetric algebraic-geometry scheme.
3. A closed form for the forbidden intermediate dimensions of a general Young permutation module M^λ , not merely $M^{(n-1,1)}$, in defining characteristic, expressed through the p -regular Specht-filtration data.
4. Whether the genus gap of width $2g$ and the submodule-lattice gap admit a unified representation-theoretic statement, or are genuinely independent constraints that must both be satisfied for a recoverable group-invariant covering scheme.

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